1. **Project Name:** Hotel and Event Management System

2. **Group Members**

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3. **About the Project**

A relational database system that brings the main operations of a hotel onto one platform: room reservations, conference and event bookings, restaurant orders, billing, and customer feedback. The business rules are held inside the database as constraints, functions, procedures and triggers, so they apply no matter how the data is reached.

The system contains sixteen tables, five views, three stored procedures, two functions, three triggers and five user roles.

4. **Technologies**

Component	**Technology**
Database management system	MariaDB 12.3

Database client	HeidiSQL
Programming language	Python 3
Web framework	Streamlit
Database driver	mysql-connector-python
Password hashing	bcrypt

5. Folder Structure

Folder: Hotel_Management_and_Events_Planning_System/

1. ***Project:*** SQL scripts
2. ***App:*** Python application
3. ***Documentation:*** Report and guides
4. ***Run.bat:*** Starts the application

How to install:

- Install these first:
    1. **Python 3** from python.org. Tick "Add Python to PATH" during installation.
    2. **MariaDB** from mariadb.org. Set a root password when asked and write it down.
    3. **HeidiSQL**, which comes with MariaDB on Windows.

Then open the ***App*** folder in File Explorer, click the address bar, type ***cmd*** and press Enter. In that terminal:

python -m venv venv

venv\Scripts\activate

pip install mysql-connector-python bcrypt streamlit

How to create and populate the database:

Open HeidiSQL and connect with hostname ***127.0.0.1***, user ***root***, port ***3306*** and your root password.

Load and run these files one at a time, in this order. Use File, then Load SQL file, then press the run button.

Order	**File**	**What it does**
1	01_hotel_ddl.sql	Creates the database and the sixteen tables
2	02_hotel_dml.sql	Loads 312 rows of sample data
3	03_app_user.sql	Creates the login table
4	04_views.sql	Creates the five views
5	05_functions.sql	Creates the two functions
6	06_procedures.sql	Creates the three stored procedures
7	07_triggers.sql	Creates the three triggers
8	09_security.sql	Creates the roles and the six database accounts

The order is not optional, because each file depends on the ones above it.

Check it worked:

```
USE hotel_db;
SELECT COUNT(*) FROM Reservation;
SELECT COUNT(*) FROM Payment;
```

You should get 25 and 16.

Two files are not part of the build. ***08_queries.sql*** holds the ten reporting queries and creates nothing. ***99_verification.sql*** is the test script and should be run only after everything else is in place, since several of its tests change data.

Note on files 5, 6, and 7. These contain ***DELIMITER*** blocks, which some versions of HeidiSQL do not handle. If they fail with a syntax error near **$$**, run them from the MariaDB command line instead:

```
"C:\Program Files\MariaDB 12.3\bin\mysql.exe" -u root -p
SOURCE C:/path/to/Project/05_functions.sql;
```

Use forward slashes in the path and no quotation marks.

6. How to Create the Login Accounts

In the ***App*** folder terminal, with the environment active:

```
python seed_users.py
```

This creates the five application accounts and stores each password as a bcrypt hash. Running it twice is safe, because it skips any username that already exists.

7. How to Run the Application

```
streamlit run streamlit_app.py
```

- A browser tab opens at ***http://localhost:8501.***
- Streamlit is never started with ***python.*** The command is always ***streamlit run.***

Alternatively, edit the two paths at the top of ***run.bat*** to match your machine and double click it.

8. Test Account Credentials

Application logins

UserName	**Password**	**Role**	**Page Available**
Admin	admin 123	Admin	All eight
Joyce	manager123	Manager	All eight
Daniel	reception123	Receptionist	Guests, Reservations, Rooms, Events
Victoria	restaurant123	Restaurant	Restaurant only
Florence	accounts123	Accountant	Invoices and Reports

Database Accounts

These are held in ***config.py*** and are never typed by the user. The application selects the matching account based on the role returned at sign in.

Account	Password
hotel_app_login	ChangeMe_Login1
hotel_app_admin	ChangeMe_Admin1
hotel_app_manager	ChangeMe_Mgr1
hotel_app_reception	ChangeMe_Rec1
hotel_app_restaurant	ChangeMe_Res1
hotel_app_accounts	ChangeMe_Acc1

The passwords in ***config.py*** must match those created by ***09_security. sql***. If they disagree, the application reports error 1045 at sign in.

9. Common Problems

"Can't connect to MySQL server" means MariaDB is not running. Press the Windows key, type Services, find MariaDB and start it.

"Access denied for user" means a password in ***config.py*** disagrees with the account in MariaDB. The message names which account.

"streamlit is not recognized" means the virtual environment was not activated. Run ***venv\Scripts\activate*** first.

"Unknown database hotel_db" means the DDL file was not run, or it failed partway. Run ***DROP DATABASE IF EXISTS hotel_db;*** and start again from file 1.

Backup and Recovery

mysqldump -u root -p --routines --triggers --events hotel_db > hotel_backup.sql

The ***--routines*** and ***--triggers*** flags are needed because mysqldump leaves stored procedures and triggers out by default. Restore with:

mysql -u root -p hotel_db < hotel_backup.sql